A Tiny Robot Swims With Living Muscles

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Most swimming robots use a motor. A new robot from MIT does not. It swims using a thin layer of living **muscle cells**.

The robot is soft and about the size of a stick of gum. It has two fins. The fins are coated with **muscle cells** thinner than a human hair. Scientists changed the cells so they squeeze when light shines on them.

When light hits one fin, that fin flaps. This pushes the robot forward. By changing which fin gets light, and how often, the team can steer the robot and change its speed.

In tests, the robot swam through an underwater maze. At its fastest, it moved about four times its body length in one minute. Ritu Raman, from MIT, said the robot is quite strong for its size.

Machines that mix living **tissue** with human-made parts are called **biohybrid** robots. This study, in the journal Advanced Functional Materials, is the first to show a very thin robot like this traveling from place to place. One day, such soft robots might do **delicate** jobs in **fragile** places.

Word Treasure

muscle cells -- Muscle cells are tiny living parts that can tighten and help things move.
tissue -- Tissue is living material made of many cells working together.
biohybrid -- Biohybrid means made by mixing living tissue with human-made parts.
delicate -- Delicate means easy to harm and needing gentle care.
fragile -- Fragile means easy to break or damage.

Background you need

Biohybrid robots mix living tissue with parts made by people.

In this robot, light makes thin muscle cells squeeze, which flaps its fins and moves it through water.

Soft robots like this could one day help with gentle jobs in places that are easy to damage.

Break it down

WHO: The MIT team that built the swimming robot
WHAT: They made a soft robot whose light-powered, muscle-coated fins let it swim and steer
WHERE: In water, where it was tested in an underwater maze
WHY: To show that a very thin biohybrid robot could travel from place to place

Why it matters

It shows kids that living cells can become part of a machine and might one day do gentle jobs in fragile places.

Test what you remember

1. What makes the new robot swim instead of a motor?

(A) A thin layer of living muscle cells
(B) Magnets under the water
(C) A motor hidden in its tail
(D) A pair of tiny wheels

2. How big is the robot?

(A) As large as a dolphin
(B) Too small to see
(C) About the size of a stick of gum
(D) The size of a shoe

3. What happens when light shines on one fin?

(A) That fin flaps
(B) Both fins fall off
(C) The fin changes color
(D) The robot stops moving

4. How can the team steer the robot and change its speed?

(A) By changing the water color
(B) By adding a bigger motor
(C) By pulling it with a string
(D) By changing which fin gets light and how often

5. What did the robot do during tests?

(A) It climbed a wall
(B) It flew over a maze
(C) It carried a camera across a lake
(D) It swam through an underwater maze

6. What are machines that mix living tissue with human-made parts called?

(A) Gum robots
(B) Muscle motors
(C) Light swimmers
(D) Biohybrid robots

Answer key (try the quiz first!): 1: A 2: C 3: A 4: D 5: D 6: D

Questions to think about

1. Ritu Raman: She said the robot is quite strong for its size.

MY THOUGHTS (at least 20 words):